Lesson 7.1 — The Unit Circle

Big idea: The unit circle defines sin and cos for ANY angle, not just acute ones. Each angle θ corresponds to a point (cos θ, sin θ) on the circle of radius 1.

Quick review

Unit circle: circle centered at the origin with radius 1.

Coordinates: a point on the unit circle at angle θ (measured from positive x-axis, counter-clockwise) is (cos θ, sin θ).

Key angles (degrees): 0°, 30°, 45°, 60°, 90°, 180°, 270°, 360°.

Key values:
cos 0° = 1, sin 0° = 0
cos 30° = $\sqrt{3}/2$, sin 30° = 1/2
cos 45° = $\sqrt{2}/2$, sin 45° = $\sqrt{2}/2$
cos 60° = 1/2, sin 60° = $\sqrt{3}/2$
cos 90° = 0, sin 90° = 1

Quadrant signs: Q1 both +; Q2 sin +, cos –; Q3 both –; Q4 sin –, cos + (mnemonic: All Students Take Calculus).

Worked example

Worked example. Find sin 150° and cos 150° using a reference angle.

150° is in Q2 with reference angle 30°. sin 150° = +sin 30° = 1/2. cos 150° = –cos 30° = $-\sqrt{3}/2$.

Warm-ups

1. What are the coordinates on the unit circle at θ = 0°?
2. Find sin 90°.
3. Find cos 180°.

Practice

1. Find sin 120° and cos 120°.
2. Find sin 225° and cos 225°.
3. Find sin 300° and cos 300°.

Challenge

1. If θ is in Q3 and sin θ = –2/3, find cos θ exactly.
2. Find all angles $0° \le \theta < 360°$ with sin θ = $\sqrt{2}/2$.

Answer key — Lesson 7.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. (1, 0).
2. 1.
3. –1.

Practice

1. Q2: $\sin 120° = \sqrt{3}/2$, $\cos 120° = -1/2$.
2. Q3: $\sin 225° = -\sqrt{2}/2$, $\cos 225° = -\sqrt{2}/2$.
3. Q4: $\sin 300° = -\sqrt{3}/2$, $\cos 300° = 1/2$.

Challenge

1. $\sin^2\theta + \cos^2\theta = 1 \Rightarrow \cos^2\theta = 1 - 4/9 = 5/9 \Rightarrow \cos\theta = \pm\sqrt{5}/3$. In Q3 cosine is negative $\Rightarrow \cos\theta = -\sqrt{5}/3$.
2. $\theta = 45°$ and $135°$ (Q1 and Q2 reference angle 45°).